

🏆 1ST PRIZE, ENTERPRISE · AI ON APPLE SYDNEY HACKATHON

CraftLens

An on-device AI coach for hands-on crafts. Skill one: Haircut Coach.

Mac and iPhone · Runs on your own devices · No cloud

a lens on any craft

Akhil Rajan

THE PROBLEM

A fresher learns the cut by watching. Nobody can watch every cut.

> “My students need real-time feedback when they practise, but I can’t be there to guide every cut.”
>
> Ash, barber with 18 years of experience

The mentor is busy

The skilled barber coaching new staff is also cutting clients. Practice on a mannequin often happens alone.

The angle is the skill

Lift a section a little under 90° and weight stacks underneath; a little over and it comes out short. It's hard to see on your own hands.

Feedback comes too late

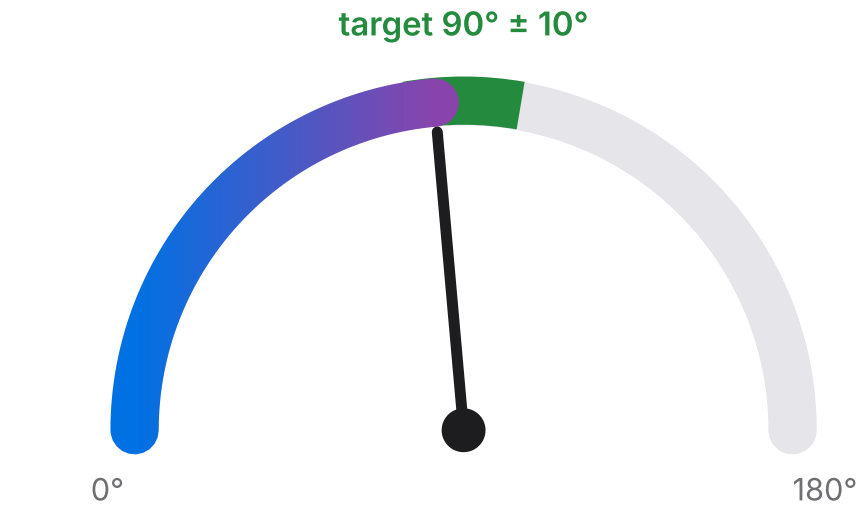
Mistakes show once the hair is cut. A coach is most useful in the moment, while the section is still in the fingers.

WHAT IT DOES

A coach that watches your hands and talks you through the cut

The iPhone on a tripod is the camera. The Mac measures the elevation angle of the holding hand live, compares it with the recipe's target and speaks a short cue. The trainee can ask questions out loud and gets a debrief when the cut ends.

First recipe: a uniform layer at 90° on a mannequin, in 6 steps, held to ±10°.



85°

Live reading · Nape guide

"A little higher."

THE JOURNEY

From a scanned head to a skill record

1
Scan
The iPhone app guides an Object Capture of the mannequin; the Mac builds the 3D head.

2
Plan
Sections and 90° guide lines drawn on that head, current step highlighted.

3
Coach
Live angle gauge and voice cues, after a quick setup check.

4
Ask
"Why ninety here?" answered out loud, from the recipe.

5
Review
Time and accuracy per step, and a spoken debrief.

6
Progress
Time on target per cut, and how each step is trending.

7
Library
The salon's own documents and shop rules, curated by the mentor.

HOW COACHING WORKS

It measures, follows along and checks, all at once

15× / s
Watching your hand

Measure. The angle you're holding the section at, compared with the step's target, with a short spoken cue when it's off. A tilted tripod is corrected for you.

every 3 s
Following along

Follow along. A fresh look at the camera to see which small move the trainee is on: taking a section, combing out, pinching, lifting, cutting.

step end
Checking the result

Check. A quick look at the step's checks, like "are the other sections clipped away?". It answers looks right, concern or can't tell, never a guess at a length.

Beginner

Says the next move if it doesn't start within a few seconds; explains how when they stall.

Practising

Less prompting. Angle cues and help only when needed.

Confident

Mostly quiet. Progress suggests when to move up a level.

BUILT TO BE TRUSTED

The AI explains. It never sets the angle.

Numbers come from measurement

The angle is measured from the camera and compared with the recipe's target. The AI only puts the measured figures into words.

It observes, it doesn't guess

Visual checks give a fixed answer. A concern is spoken in the salon's own words, and only once those checks have been proven on real cuts.

Answers stay in the recipe and the library

One or two spoken sentences from the recipe or mentor-approved documents, with sources shown. Otherwise: ask your educator.

The coach keeps coaching

Asking a question never pauses the coaching. If Apple Intelligence is off or still downloading, the coach says so and carries on.

WHAT YOU NEED

Hardware a salon may already have

A Mac

Apple silicon, with Apple Intelligence turned on. It runs the coach.

An iPhone on a tripod

The coaching camera and microphone, beside the mannequin.

A Pro iPhone to scan

Any iPhone with LiDAR (12 Pro or later) scans the head. A folder of photos also works.

A mannequin head

Plus scissors, a cutting comb and sectioning clips.

No subscriptions, accounts or internet connection needed for coaching. AirPods connected to the Mac help in a noisy salon.

PRIVATE BY DESIGN

Client faces never leave the salon's own devices

Nothing goes to the cloud and there are no accounts. The only network traffic is between the trainee's iPhone and Mac, on the local network, encrypted, and only while it's being used.

Local, encrypted link

The iPhone's video and microphone reach the Mac encrypted, paired once with a 4-digit code.

AI that runs on the Mac

Camera photos are looked at on the Mac only, never sent to any cloud service.

Works offline, costs nothing per use

Every answer and check runs on the Mac, so there are no usage fees and no internet needed. Check photos are kept only if you turn that on.

COACHING FOLLOWS A RECIPE

Every cut is a step-by-step recipe

Each recipe holds the steps, the target angle and how much leeway is allowed, the small moves in each step, what to check, and the words the coach uses. The salon's mentor shapes how it's taught.

UNIFORM LAYER AT 90°

1 Section the head	3 moves
2 Nape guide	90° ± 10°
3 Back of the head	90° ± 10°
4 Sides	90° ± 10°
5 Top and crown	90° ± 10°
6 Cross-check	90° ± 10°

"Lift a little higher. Straight out from the head."

BEYOND BARBERING

Barbering is skill one. The same coach can teach any hands-on skill.

Watch what you do

The camera follows your hands and your tools as you work.

Compare with an expert

Each step is checked against how a skilled mentor does it.

Coach in the moment

A spoken cue while you can still fix it, not after.

Trades

Tool angles, order of steps, safe technique.

Cooking

Knife work and technique, step by step.

Painting

Brush angle, strokes and sequence.

Instruments

Hand position and posture while you play.

Each new skill is a new recipe, not a new app: an expert maps the steps and what good looks like, and the coach does the rest.

WHERE IT IS

Built in 44 hours. Next: prove it on real cuts.

Built

- Idea to working app in 44 hours, 1st prize in the Enterprise category
- Mac coaching app and the CraftLens Scan iPhone app
- The iPhone as a wireless camera and microphone for the Mac
- Guided coaching, spoken questions, debrief and progress

Next

- Validate the angle against the ±10° goal with a mentor
- Score the visual checks on real stills before they speak
- Pilot with a salon's freshers and their mentor
- More cuts, then other hands-on skills



a lens on any craft

Haircut Coach is the first skill. Any hands-on skill can be next.

Training new barbers? Let's run a pilot with your mentor and freshers.

Built at the AI On Apple Sydney Hackathon. 1st prize, Enterprise category.

linkedin.com/in/akhilpeterrajan